

A detailed technical diagram of an air compressor system is shown in the background. It features a central 3D wireframe model of the compressor unit, which includes a large electric motor and a complex arrangement of pipes, valves, and pressure vessels. Surrounding this central model are various 2D schematic diagrams. In the top left, there is a 'FLOWCHART' showing the sequence of operations for 'PRIMARY 1', 'SECONDARY 1', and 'TANKING 11'. To the right of the main unit, there are 'VALVE POSITION' indicators for 'DEF 1', 'DEF 2', and 'DEF 3'. Further right, a 'BLEEDING TEST' section includes a pressure gauge and a 'TEST' button. At the bottom right, a 'TEST LOG' table is visible, with columns for 'TEST DATE', 'TEST TIME', and 'TEST RESULT'. The entire diagram is overlaid with a glowing blue grid and lines, giving it a high-tech, digital appearance.

NK Starting Air Compressors

Technical Training & Operational Mastery (Modules 1-3)

PT. **ITI** MARINE & OILFIELD UTAMA



MODULE 1: Platform & Provenance

Corporate pedigree, identification, and system architecture.

STATUS: ACTIVE >>



MODULE 2: Kinematics & Internal Mechanics

X-ray diagnostics: 19-point anatomy, rotational drives, and stepped piston thermodynamics.

STATUS: ACTIVE >>



MODULE 3: Operations, Safety & HMI

Sensory checks, safety fail-safes, and digital HMI maintenance protocols.

STATUS: ACTIVE >>

The Provenance of Engineering: The Neuenhauser Group



The Footprint

Employs around
2,800 people.



Encompasses over 25
companies globally.

Production sites spanning
Germany, USA, Hungary,
and the Czech Republic.



The Pedigree

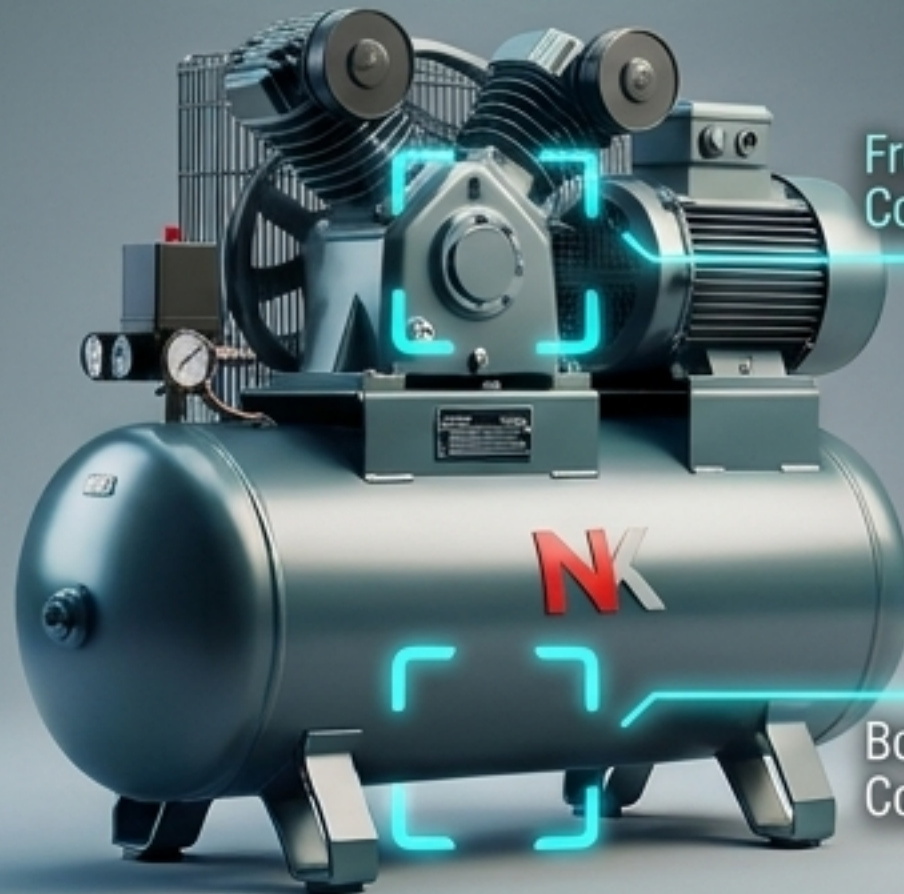
■ Made
■ in
■ Germany



neuenhauser[®]
GRUPPE

NK (Neuenhauser Kompressorenbau) is the dedicated compressor division, established in 1986 from the Jos. L. Meyer legacy.

Diagnostic Targeting: Nameplate Decoding



Front or Side of
Compressor Block

Bottom of
Compressor Block

NK	AIR COMPRESSOR		
	NK COMPRESSOR		
Model Number	NK-93903		
Serial Number	123485677899		
Year of Manufacture	2024		
Pressure Rating	R00	kvd	mm
Power Details	500		
Power Details	150V	No	20A
Prepared in NK Empresor			

Model Number
(Model Series & Suffixes)

Serial Number

Year of Manufacture

Pressure Rating
(e.g., 30 bar)

Power Details
(Voltage/Current requirements)

System Architecture Matrix: 2-Stage vs. 3-Stage



2-Stage Compressor



Specs: Small & Compact Unit. 230V, min 20A supply.

Application: Intermittent use (e.g., factories, textiles).

Cooling: Recommended for Air-Cooled configurations (open spaces with free breather air).

3-Stage Compressor



Specs: Large, High-Efficiency Units. Allows for smaller tanks.

Application: Heavy duty (marine, aviation, power plants).

Cooling: Recommended for Water-Cooled configurations to combat high ambient temperatures (e.g., ship engine rooms at 42°C - 50°C).

Transferring Power with Micro-Precision

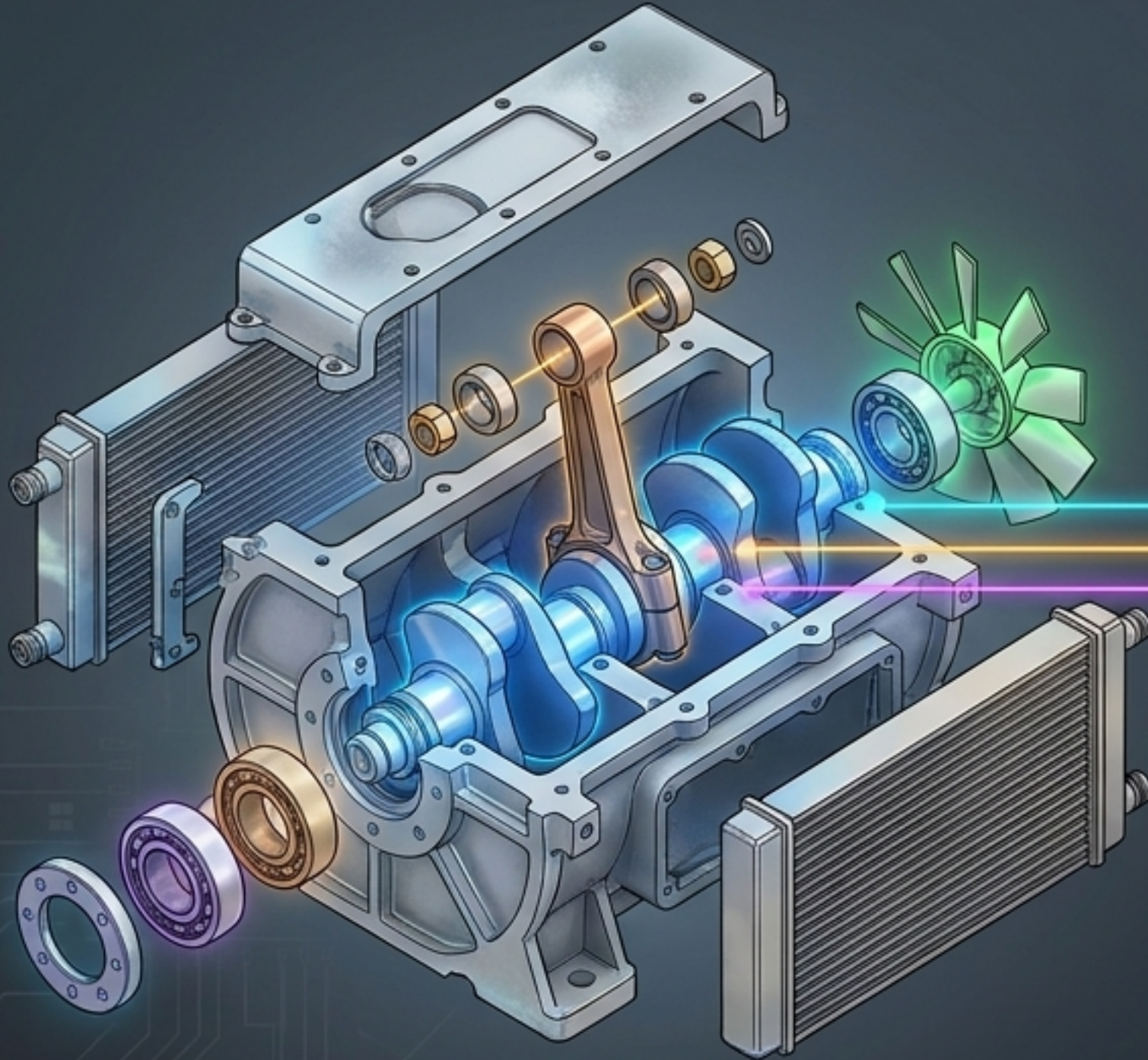


1 Crankshaft Assembly
Anchors the rotational drive.

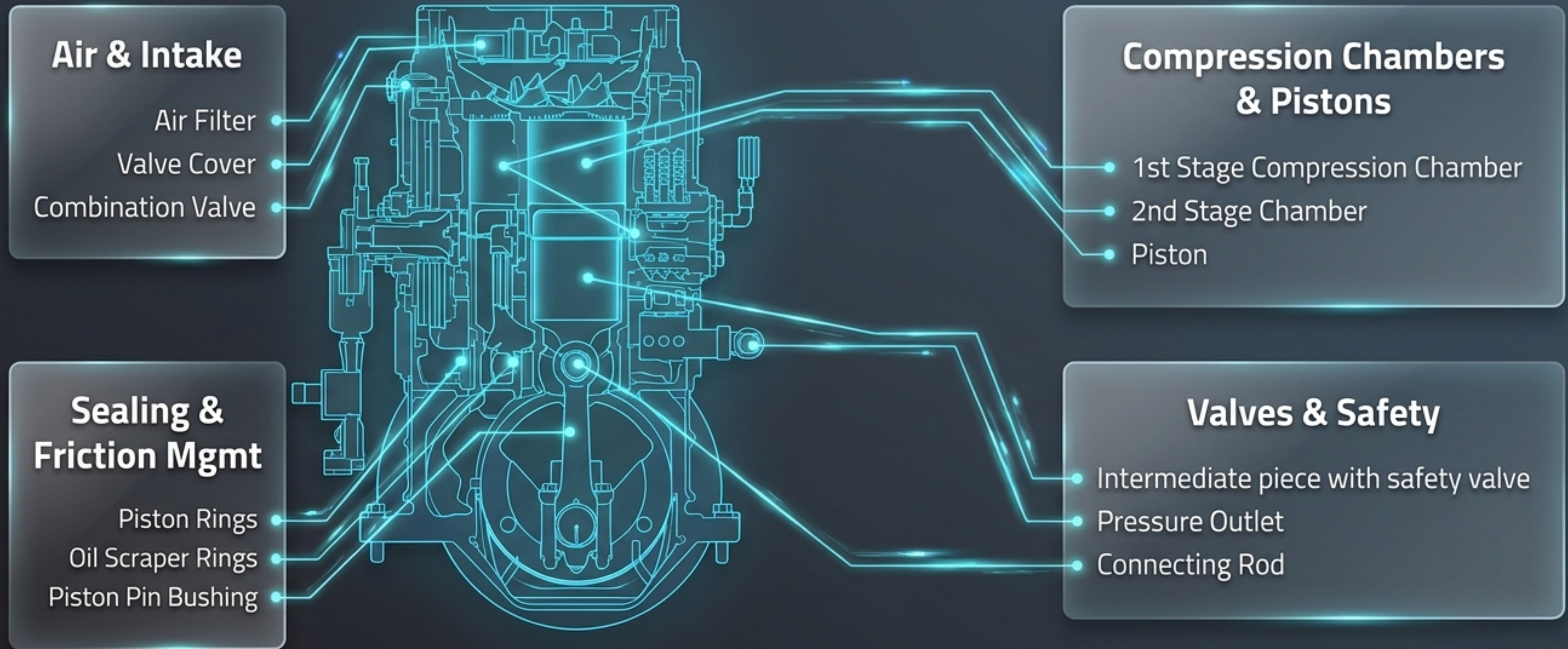
2 Connecting Rod (Pleuelstange)
The critical linkage translating rotation to vertical piston thrust. Includes precision hexagonal nuts and pins.

3 Bearing Systems
Connection Rod Bearing (Pleuellager) and Needle Bearing (Nadellager).

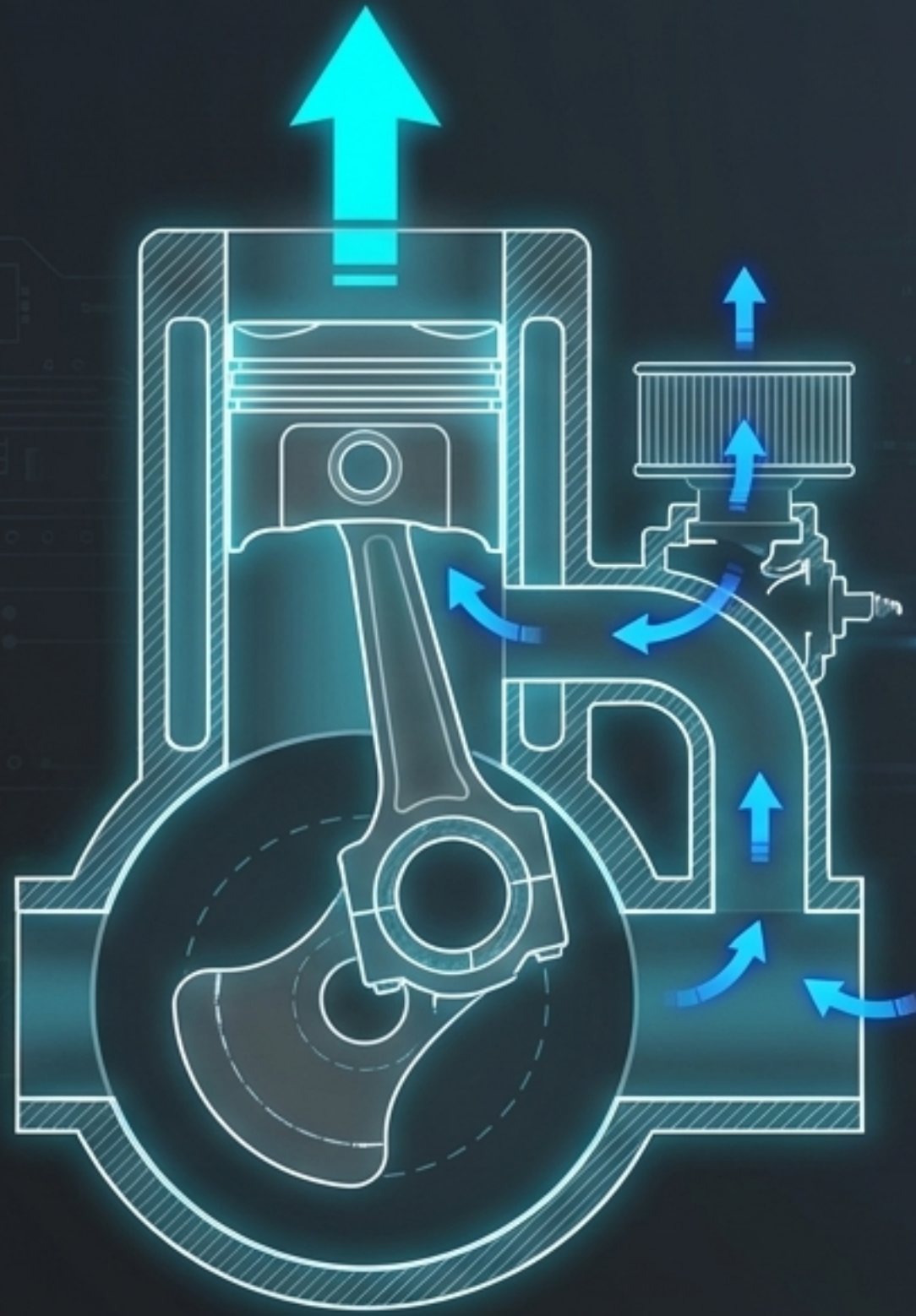
Crucial Insight: Engineered to withstand continuous high-friction operation during rapid pressure transitions.



X-Ray Diagnostic: The 19-Point Anatomy



Kinematics Phase 1: The Upward Stroke (LP)



Physics Telemetry Panel

Action:

Piston moves upward. Ambient air is drawn through the intake air filter and combination valve.

Stage 1 Status (Active Compression):

Air is squeezed in the 1st stage compression chamber.

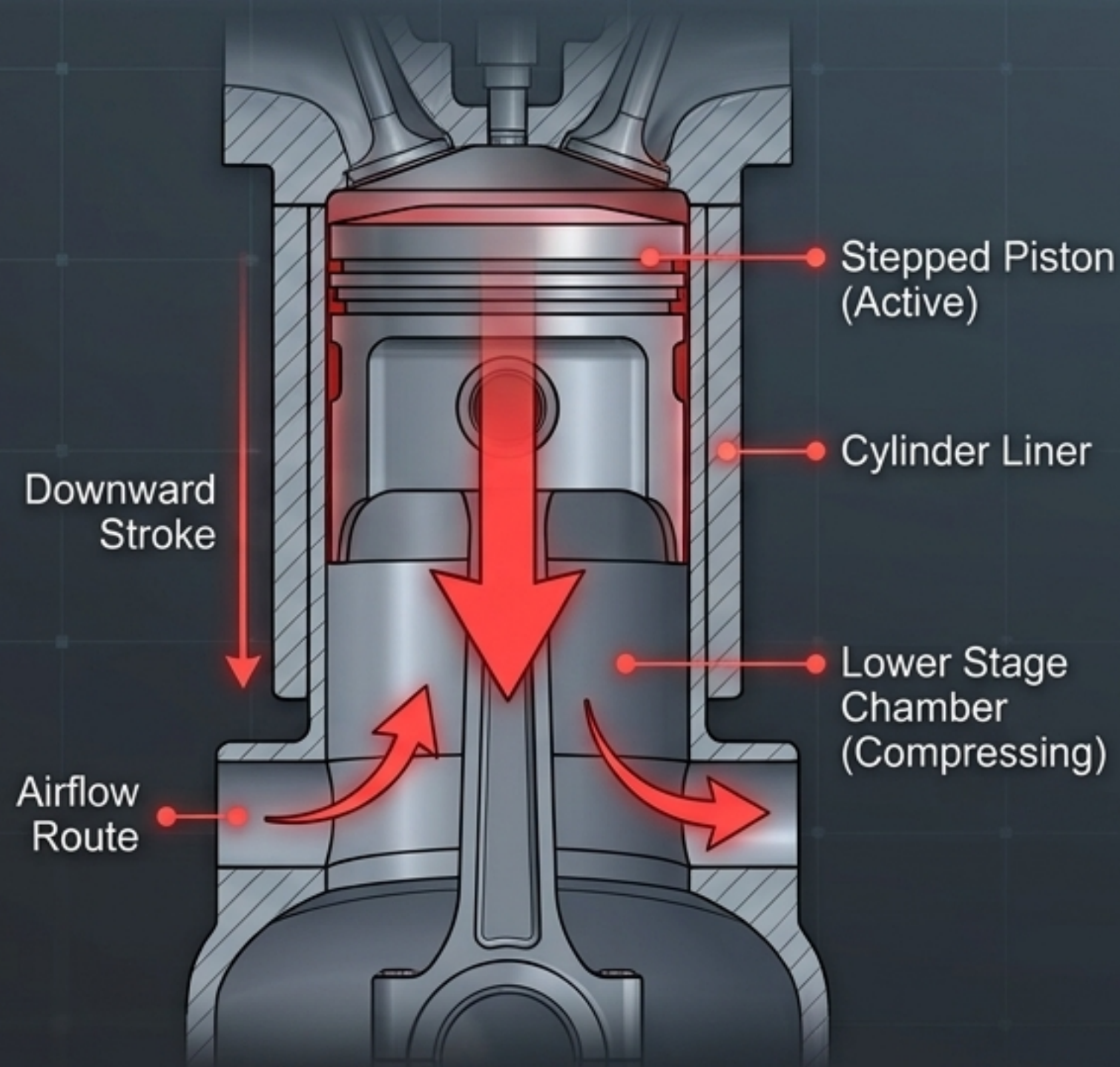
Live Pressure Indicator:

↑ [5.6 bar] ↑

Intercooling Routing:

Compressed air then passes through the intercooler toward the suction valve of the 2nd stage.

Kinematics Phase 2: The Downward Stroke (HP)



Physics Telemetry Panel

Action:

Piston moves downward.



Stage 2 Status (Active Compression):

The pre-compressed 5.6 bar air is aggressively squeezed in the 2nd stage chamber, increasing pressure to a final 30 bar.

Live Pressure Indicator:



[**30 bar**]



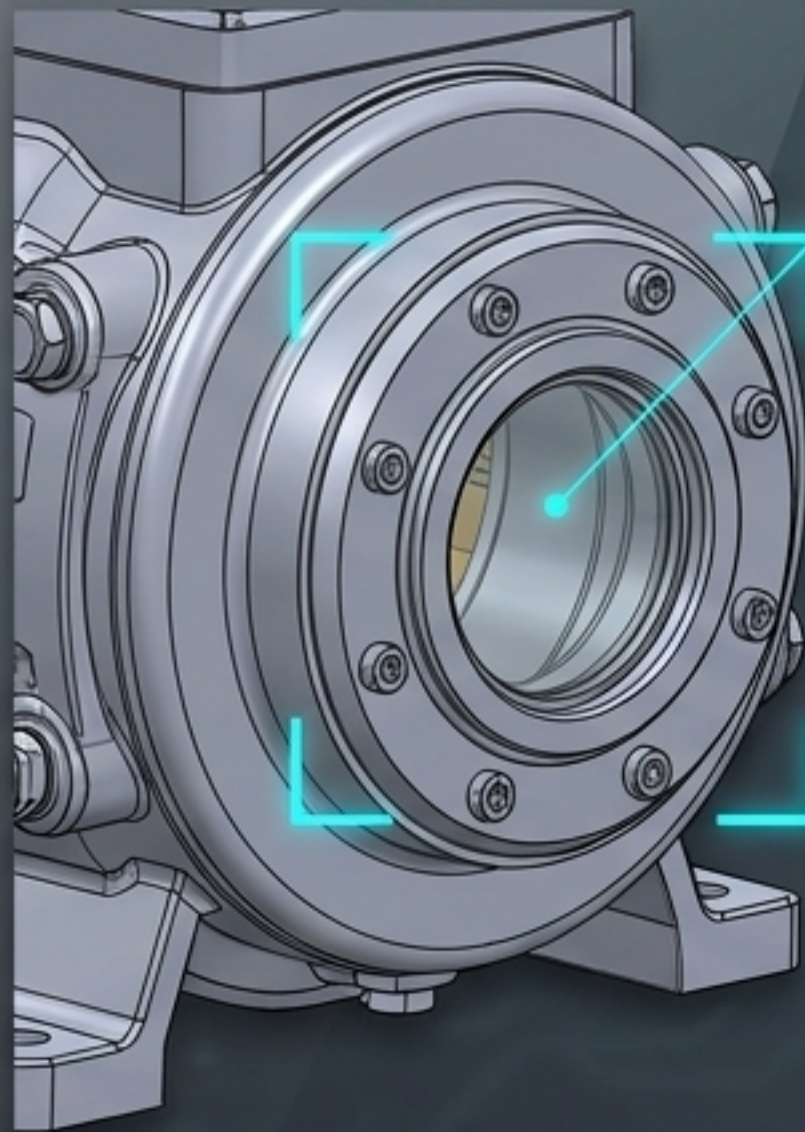
Stage 1 Status (Simultaneous Suction):

As the piston descends, it creates a vacuum in the top chamber, sucking fresh ambient air through the combination valve into the 1st stage to restart the cycle.

ROUTE: FRESH AIR IN →

Physical Telemetry: Daily Sensory Diagnostics

The Golden Viewport



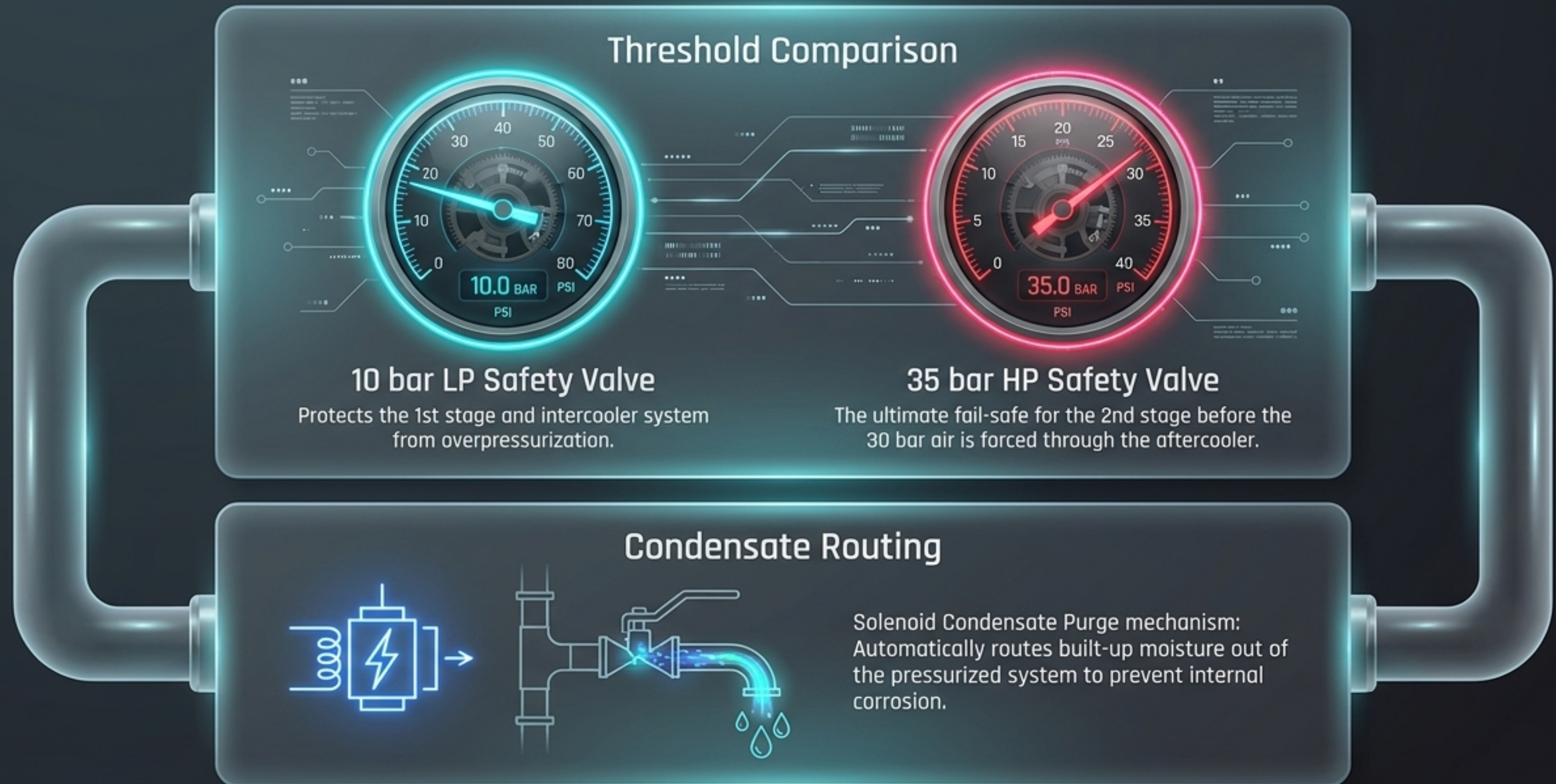
Monitor for mechanical health and safe vibration tolerances during operation.

The Oil Sight Glass



Observe oil color, clarity, and meniscus level. Ensures needle bearings and connecting rods are properly lubricated for the intense 30-bar cycles.

Fail-Safe Architecture: Valves & Purge Systems



HMI Air Navigator: Operational Monitoring



HMI Air Navigator: Security & Maintenance Protocol





PT. **ITI** MARINE &
OILFIELD UTAMA

Maximum Performance Under High Pressure.

Through rigorous daily diagnostics, mastery of stepped-piston thermodynamics, and strict adherence to HMI protocols, we ensure total operational readiness.

Technical Training Modules 1-3 Completed. For advanced diagnostics, consult your PT. ITI engineering lead.